

ADOS[®]-2 Training for Clinicians

Toddler Module Introductory Workshop, February 24, 2025
A Webinar by The WPS Professional Learning Center (ProLearn™)

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Learning Objectives

1. Distinguish the key features of the ADOS-2 Toddler Module approach, including the rationale for the approach and how it enhances autism assessment in toddlers;
2. Clarify use of the ADOS-2 for operationalizing diagnostic criteria for toddlers with autism spectrum disorder;
3. Distinguish administration procedures for the ADOS-2 Toddler Module;
4. Demonstrate how to apply ADOS-2 Toddler Module codes;
5. Demonstrate how to score ADOS-2 Toddler Module algorithms; and
6. Interpret the basic psychometric support for the ADOS-2 Toddler Module.

This training presumes prior completion of the ADOS-2 Clinical Workshop or ADOS-2 Training Video Program.

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Disclosures

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Experience Needed to Use the Toddler Module

Clinical Use	Research Use
Previous clinical training and experience with Modules 1–4 of the ADOS-2	Previous clinical training and experience with Modules 1–4 of the ADOS-2
ADOS-2 Toddler Module Course	ADOS-2 Toddler Module Course
	Establish ADOS-2 research reliability: •Modules T, 1, 2 and •Modules 3, 4 (Separate reliability is not needed specifically for Toddler Module.)
Prior assessment experience with toddlers and knowledge of infant and early development	

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What This Training Is Not

- Equivalent to full preparation for clinical or research use
- Training on general assessment of toddlers
- A replacement for reading the Manual!

Further practice and experience will be needed before you can use the ADOS-2 Toddler Module reliably for clinical or research purposes.

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Identifying Autism Spectrum Disorder (ASD) in Toddlers

- Early detection with subsequent early intervention can lead to substantially better prognosis, including
 - Improved nonverbal problem solving;
 - Improved expressive and receptive language abilities;
 - Better adaptive functioning; and
 - Fewer maladaptive behaviors.

Source: Kleinman, J. M., et al., 2008.

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Emergence of ASD Symptoms in Toddlers

- Symptoms generally emerge between the first and second years of life
 - Although some symptoms may be present during infancy
- Parents generally report a concern about their child's development or behavior before the child's second birthday
 - 17 months—average age that parents first report a concern to their pediatrician

Source: Kleinman, J. M., et al., 2008; Mandell, D. S., et al., 2005.

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Emergence of ASD Symptoms in Toddlers

(cont.)



ASD in first year:
behavioral markers
at 12 months

- Atypical eye contact
- Visual tracking
- Orienting to name*
- Imitation, social smiling
- Reactivity
- Social interest
- Sensory-oriented behaviors
- Less pointing
- Less showing
- Less frequent looking at others' faces*

*Differentiated ASD from developmental delay

Source: Wetherby et al., 2004; Zwaigenbaum et al., 2005.

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Emergence of ASD Symptoms in Toddlers (cont.)



- “Regressive” onset previously thought to occur in 20%–30% of cases
 - Regression: Typical development during the first years of life, followed by period of loss in language and/or social–communication skills
- Rather than obvious loss, many children may “plateau” or fail to progress in skill acquisition
 - Prospective studies suggest this may be the most common onset pattern
- Many children show abnormalities prior to “regression”

Source: Kim et al., 2018; Ozonoff et al., 2015, 2018; Thurm et al., 2017.

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Are diagnoses of ASD in young children accurate over time?

- Stability of early diagnosis is generally high.
 - Clinical judgment combined with an ADOS-2 have accurately identified children under 24 months of age with ASD.
 - Diagnoses have been stable as children enter elementary school.
 - Some children “move off the spectrum,” but most often they continue to show residual language and/or other neurodevelopmental difficulties.
- Several recent studies have found that in most cases there are signs of disruption in development of communication and joint attention by 14 months of age.

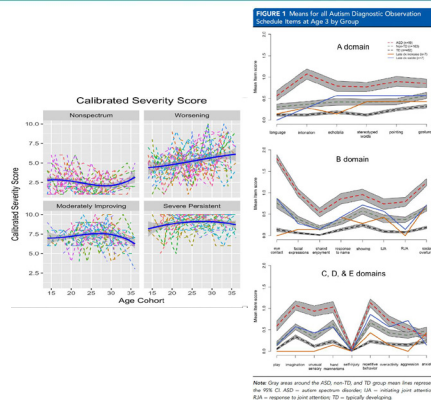
Source: Chawarska et al., 2007; Guthrie et al., 2013; Kim et al., 2015; Kleinman, J. M., et al., 2008; Landa et al., 2007; Luyster et al., 2009; Ventola et al., 2006.

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Diagnoses Are Less Stable in Some Toddlers

- If a very young toddler (e.g., 12–24 months) does not meet criteria for autism on first assessment, it does not mean the child is “cleared” for an autism diagnosis forever. → They will require repeat assessments and close monitoring.
- “Later” onset: Younger siblings diagnosed at age 5; at age 3, looked significantly different on ADOS-2 than the diagnosed group and fell somewhere between children with autism and typically developing children on most measures.



Source: Kim et al., 2018; Ozonoff et al. 2018.

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Symptom Trajectories in the First 18 Months

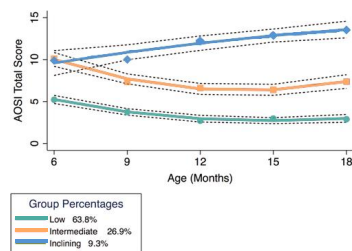
- Group-based modeling using Autism Observation Scale for Infants (AOSI) scores identified three distinct trajectories (Figure 1).
- ASD diagnosis at age 3 was **most likely** in the **inclining** group, followed by the **intermediate** group.
- However, children with ASD were distributed across all three trajectories.
 - Membership in the inclining group was **highly specific** (94.5%) but **poorly sensitive** (28.5%) to ASD.

Source: Kleinman, J. M., et al., 2008; Mandell, D. S., Novak, M. M., & Zubritsky, C. D., 2005.

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Symptom Trajectories in the First 18 Months (cont.)



- **Low** (n = 435, 64.3%) characterized by a low level and stable evolution of ASD signs
- **Intermediate** (n = 180, 26.6%) had intermediate and stable levels
- **Inclining** (n = 61, 9.3%) had higher and progressively elevated levels of ASD signs

Source: Zwaigenbaum et al., 2021.

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Siblings of Children With ASD

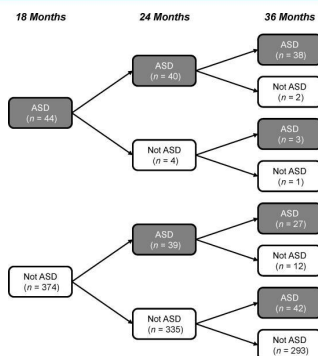
- Baby siblings of children with ASD
 - Recurrence rate is high (15%–25%; Ozonoff et al., 2011).
 - Up to 28% of high-risk infants without a later ASD diagnosis exhibit some ASD symptoms at 12 months (Georgiades et al., 2013; Macari et al., 2012; Ozonoff et al., 2014).
 - Symptoms in a small subset of children may not be fully manifested until 3–5 years of age (e.g., Rogers et al., 2018).

Source: Kleinman, J. M., et al., 2008; Mandell, D. S., et al., 2005.

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Infant Sibling Study's Diagnostic Process—Stability and Change



Take-home:
Even expert diagnosticians are often not comfortable diagnosing ASD until 36 months.

Source: Ozonoff et al. (2015). *J Child Psychol Psychiatry*, 56, 988–998.

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Challenges in Assessing Toddlers

- Manifestations of core symptoms of autism are different depending on:
 - Age
 - Language level
 - Nonverbal cognitive abilities
- Early “red flags” can be very subtle.
 - Parents may not be aware that an aspect of their child’s behavior is a symptom of autism.
 - Symptoms can be easily dismissed and attributed to normal variations in development (e.g., “Dad was just like this at 2”).

Accurate diagnosis is dependent on **comprehensive** assessment by **trained, experienced** clinicians with **knowledge of infant and early development**.

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Challenges in Assessing Toddlers (cont.)

- Assessing toddlers requires different strategies and flexibility.
- Examiners must feel comfortable with both toddlers and their parents.
- The examiner's behavior is central to a successful administration.
 - The examiner's behavior matters because the examiner must create a fun social context for the toddler.
 - The ADOS-2 Toddler Module should not feel like a "test" but must follow the standardized protocol—this can be a challenging endeavor for an examiner who is new to the ADOS-2 and/or unfamiliar with toddlers.

Source: Thurm et al., 2017; Ozonoff et al., 2015; Kim et al., 2018; Ozonoff et al., 2018.

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The ADOS-2 Toddler Module

- Same basic structure as the other ADOS-2 modules
- Appealing toys and simple routines
- Examiner provides more structure than in other modules
- Less focus on initiations by the child and greater focus on responses



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Toddler Module Is Standardized By...

- Activities
- Materials
- Behavior of the examiner
- Behaviors to be observed
- How the child's behavior is quantified
- Algorithm selection
- Risk ranges
- Training of the examiner
- Standards for achieving and maintaining reliability

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When to Use ADOS-2 Toddler Module

Chronological age	12–30 months If 31 months or older (no speech to phrase speech) → Module 1
Language	Nonverbal to primarily single words and occasional phrases If flexible 3-unit phrases → Module 2 (regardless of age!)
Nonverbal mental age	MUST have nonverbal mental age at least 12 months
Motor	MUST be walking
If the toddler is not walking and/or has a nonverbal mental age of less than 12 months , the validity of the Toddler Module is weakened ; risk of over-classification	

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BEFORE the Toddler Module Administration

- Prepare room and materials before participant arrives.
- Arrange the environment:
 - Prepare a toddler-friendly room with toddler-size table and chairs.
 - Do not use a child-sized chair with restraints; child must be able to move in and out of chair independently.
 - Arrange task materials accessibly, but out of child's reach.
 - Provide comfortable chair for parent/caregiver.
- Proper materials:
 - Make sure your equipment works (e.g., bubble toy, remote car or bunny, balloons).
 - Have a stash of extras, including batteries.
 - Have the proper snack materials (obtain information about special diets/eating restrictions prior to administration).

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BEFORE the Toddler Module Administration (cont.)



- Proper materials (continued)
 - Snacks and cup should be appropriate for the toddler's age:
 - Use a disposable plastic sippy instead of open paper cup.
 - Offer dissolvable solids instead of hard snacks (e.g., puffs instead of pretzels).
- Tell the parent/caregiver what to expect and how to behave.
- Observe the child's comfort level and respond accordingly.
 - Provide time to warm up.
 - Kneel or sit at child's level.
 - Avoid starting with the child in caregiver's lap.
 - Avoid introducing personal objects (e.g., pacifier, blanket, favorite toy, etc.).

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What a Toddler Testing Session Might Look Like

- Warm-up period: Make sure that the child is comfortable with you and the space
- Direct testing to establish developmental level (e.g., Mullen Scales of Early Learning, Bayley Scales of Infant Development)
- Break to set up your ADOS-2 materials
- ADOS-2 Toddler Module



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DURING the Toddler Module Administration

- Aim for standardized administration of 40–60 minutes, completed in one sitting.
- There should be only **one** examiner.
 - An observer may help with materials, but codes should be assigned only by the examiner (and/or parent/caregiver as specified).
- Be flexible! Match your style to the child.
- Observe and promote interactions between child and caregiver as necessary, BUT
 - Encourage parent/caregiver to allow child to participate independently.
 - End the session on a positive note.

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AFTER the Toddler Module Administration

- Assign codes *immediately* following the assessment.
 - Codes should ALWAYS be assigned based on live (not taped) administration, and only on behaviors that occurred DURING the ADOS-2.
 - Importantly, psychometric data are based on information from live coding.
 - Video recordings can be used for detailed study, assessment of reliability, etc.
- Use information from the Toddler Module in combination with all other information from the diagnostic assessment.

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Toddler Module Activities

- | | |
|--|---------------------------------------|
| 1. Free Play | 6. Anticipation of a Social Routine |
| 1a. Free Play—Ball | 7. Response to Joint Attention |
| 2. Blocking Toy Play | 8. Responsive Social Smile |
| 3. Response to Name | 9. Bath Time |
| 4. Bubble Play | 9a. Bath Time—Ignore |
| 4a. Bubble Play—Teasing Toy Play | 10. Functional and Symbolic Imitation |
| 5. Anticipation of a Routine With Objects | 11. Snack |
| 5a. Anticipation of a Routine With Objects—Unable Toy Play | |

Activities unique to the Toddler Module shown in teal.

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Free Play Materials



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Free Play Instructions

- Tell parent/caregiver to give child a few minutes to explore and play independently; observe and note child's behaviors and vocalizations.
- Join child's play and direct attention to toys. Model play, but only after providing child opportunity to do so first.
- Model pretend play after putting away cause-and-effect toys.
- When appropriate, administer:
 - Free Play—Ball
 - Free Play—Presentation of three objects
 - Blocking Toy Play
 - Response to Name

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Free Play—Ball

Instructions

- Present ball to child by bouncing on ground, rolling to child, or tossing gently into their arms.
- Encourage a simple reciprocal routine.



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Free Play—Presentation of Three Objects

Instructions

- When child spontaneously stations themselves at table, draw their attention to three objects: metal lid, plastic ring, and cylindrical tube.
- Hand each of the three objects to the child one after the other, allowing 15–30 seconds of object exploration.
- All three objects should be presented, regardless of child's behaviors with each object.
- Observe child's exploration of objects, particularly whether it is repetitive.

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Blocking Toy Play



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Blocking Toy Play (cont.)

Instructions

- When child is independently engaged in a routine with a toy (e.g., shape sorter, multiple pop-up, jack-in-the-box), limit access to the toy by covering part of it with your hand.
- Wait 5–7 seconds for the child to respond.
- Once the child acknowledges your “blocking,” remove your hand and allow the child to continue playing for 10–15 seconds.
- Repeat press once more for a total of TWO attempts.
- Code the behavior immediately in your protocol in the designated space.

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Response to Name

Instructions

- Administer when child is involved with a toy or activity...
BUT, not so overly engaged they are unlikely to respond.
- Position yourself at a distance of 3–5 feet from the child in a manner so that they have to turn their head or look up deliberately in order to see your face.
- Call child's name up to FOUR times.
 - If they do not respond, proceed to next step.

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Response to Name (cont.)

Instructions

- Ask parent/caregiver to call child's name up to TWO times.
 - If child does not respond, proceed to the next step.
- Ask parent/caregiver, "Is there any other vocalization you can use, without touching them, to get them to look at you?"
 - If child does not respond, proceed to the next step.
- Encourage parent/caregiver to do whatever they would do at home to get child's attention.

Important note: Be sensitive to the parent/caregiver during this task.

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Bubble Play



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Bubble Play (cont.)

Instructions

- Tell the parent that you want to see if the child will notice the bubbles without an adult pointing them out.
- Begin blowing bubbles with the bubble gun and wait to see whether the child notices them.
- If the child appears unresponsive, model some fun behaviors to help get the child engaged, such as popping or stomping on the bubbles while vocalizing "Pop! Pop!"
- Set up a routine with the bubbles. Give the child an opportunity to request more bubbles by pausing and waiting for them to initiate a request.
 - Maintain control (at least initially) of the bubble juice to provide greater opportunity for requesting.

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Bubble Play—Teasing Toy Play

Instructions

- Midway through “Bubble Play,” hold out the bubble toy and offer the child a turn.
- Withdraw the bubble toy as the child reaches for it, while making a playful, nonsense sound (e.g., “**Oops!**” or “**Uh-oh!**”) and smile.
- Observe the child’s response to your action, particularly their use of eye gaze (e.g., toward examiner’s face or toy).
- Repeat immediately for a total of TWO attempts. After the second press, say consolingly, “I’m sorry! I was just teasing,” give the bubble gun to the child, and allow them to play with it briefly.

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Anticipation of a Routine With Objects



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Anticipation of a Routine With Objects (cont.)



Instructions

- Slowly blow up a large balloon.* Pinching the neck of the balloon so that it won’t deflate, hold it directly in front of the child.
- Alerting the child before you do so (e.g., “Ready, set, go!” or “One, two, three!”), let go of the balloon so that the air releases from it and the balloon flies around the room.
- Wait for the child to bring the balloon to you or to indicate in some other way that they want it to be blown up again. If the child does not get the balloon, you should get it.
- Repeat the procedure a few more times, waiting each time for the child to initiate the routine.
- **Do not allow the child to attempt to blow up the balloon or place it in or near their mouth.**

**If balloons are prohibited, establish the routine with the foam dart rocket.*

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Anticipation of a Routine With Objects—Unable Toy Play

Instructions

- Administer while the child is still interested in the balloon routine.
- Put the balloon to your mouth and appear as if you are trying to inflate it. Make a puzzled expression and say, “It’s not working!”
- Repeat once more for a total of TWO attempts.

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Anticipation of a Social Routine



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Anticipation of a Social Routine (cont.)

Instructions

- Peekaboo: Hold the blanket up between your face and the child. Say, "Where's [child's name]?" Pull the blanket down with excitement, saying "Peekaboo!" or the child's name. Repeat once, hesitating with the blanket held up. If the child pulls the blanket down, repeat, then pause and hold the blanket out toward the child. Watch for a move to put the blanket on your face, on their face, or to look at you, waiting for you to repeat. If not, try once more, then proceed to tickling and swinging.
- Tickling: Say, "Here comes the tickle bug!" Hold your hands up and move them in a tickling motion while moving toward the child. Gently tickle their leg or stomach.
- Swinging: Approach with arms out and "jump" the child up and down lightly, counting down before each jump. On the second approach, wait before jumping.

*If the child does not respond positively, ask the caregiver to try the same games. If the child still does not respond positively, ask the caregiver to demonstrate a social routine familiar at home.

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Response to Joint Attention



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Response to Joint Attention (cont.)

Instructions

- During room setup, hide the remote-controlled toy under a blanket. Discreetly remove the blanket when ready to administer this activity.
- Positioning: Make a triangle in space, with you and the child facing each other and the toy at a distance to the side.
- Call child's name and/or touch them to gain attention. Up to five attempts should be made to attract child's attention. (Proceed even if you do not get the child's attention.)
- Say: "[Child's name], look! Look at that!" (exaggerating your gaze shift) up to THREE times.
- Say: "[Child's name], look at that!" (with a gaze shift and a point) up to TWO times.
- Activate the toy and then engage the child in playing with the toy for an appropriate amount of time.

Remember: "3-2-Touch"

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Responsive Social Smile

Instructions

- Ask the parent/caregiver, "Can you show me how you get them to smile, without touching them?"
- Any visual and/or vocal means may be employed at first, as long as touching is not utilized. If this is unsuccessful, encourage the parent/caregiver to touch the child to evoke a smile.

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Bath Time



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Bath Time (cont.)



Instructions

- Carry out the activity at a relatively slow pace.
- Pause during the routine to give the child a chance to exhibit spontaneous pretend play. Proceed to modeling if the child does not spontaneously pretend.
- With animation, say "Look! Here's the baby and it's time for the baby's bath! Let's give the baby a bath!"
- Pretend to turn on the water, "Let's get some water!" Test the water and say, "Oh careful! Hot!"
- Adjust the water and say, "Oh look, here's the water. Splash, splash!"
- Get baby ready for bath and say, "Let's get the baby ready for bath. Let's take her clothes off."
- Suggest putting baby in the bath. "I think the baby is ready for bath. [Pause.] Can you put the baby in?"
- Allow the child to engage in pretend play.
- Sing a common bath time song (e.g., "Rubber Duckie").

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Bath Time (cont.)



Instructions

- Prompt the child to wash the baby.
- Wash around the baby's eyes, "Oh no! We got soap in the baby's eyes!"
- Manipulate the doll as an independent agent of action (e.g., make the doll splash or wash her own hair).
- Begin to wrap up the activity and say, "Time to get out!"
- ADMINISTER THE "IGNORE" ACTIVITY.
- Put the blanket within the child's reach, "Oh, the baby is tired.... It's time for the baby to go to sleep."

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Bath Time—Ignore

Instructions

- Administer after the Bath Time routine is mostly completed.
- Leave out all the materials and explain to the parent/caregiver:
 - "We want to see what [child's name] will do when we don't pay attention to them. We won't have it last longer than a minute or so, and if they try to get your attention, you can respond as you normally would. I'll start timing now."
- Move a short distance away from the child (at least a couple feet) and begin timing the 60-second period.
 - Discretely observe the child's behavior.
 - If the child makes a clear bid for attention, using eye contact and/or vocalization, terminate the activity and check the time to note the duration of the time ignored.
- Prompt the child to continue with the Bath Time activity.

Note: If the child engages in potentially hazardous behaviors (e.g., climbing on a table or banging their head), stop the activity and re-administer at a later point.

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Functional and Symbolic Imitation



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Functional and Symbolic Imitation (cont.)



Instructions

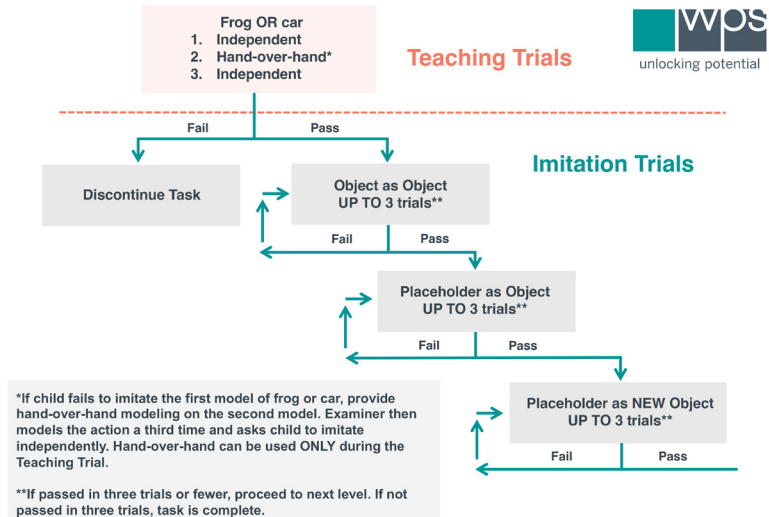
For each trial, demonstrate an appropriate action and sound effect and then ask the child to imitate. For example, say, "It's a frog. Ribbit! Ribbit! YOU do it!"

- Teaching trial(s): Real object (frog OR car, not both) UP TO three trials total (one trial hand-over-hand)
- Imitation trials, Phase 1: Real object UP TO three trials per object; use only objects not used in teaching trials
- Imitation trials, Phase 2: Placeholder representing object just demonstrated in Phase 1: UP TO three trials per object
- Imitation trials, Phase 3: Placeholder representing object not previously demonstrated; UP TO three trials per object

Note: Once all of the objects have been used, the child is "out of chances" and the activity is complete.

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Snack



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Snack (cont.)

Instructions

- Say "It's time for a snack" while placing one of each type of snack on plate in front of child. Say, "We have cookies and crackers/pretzels."
- Pause to observe whether child spontaneously requests more snack after they have eaten the food. If no request is made, hold up each container and ask, "Do you want more?"

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Toddler Module Codes

- Some items consider a single skill
e.g., Gestures, Eye Contact, Giving
- Some items consider a particular event
e.g., Response to Name
- Some items are summary codes
e.g., Quality of Social Overtures
- Summary codes take into account behaviors already considered in other codes
e.g., showing a toy or using the examiner's hand as a tool to request are types of social overtures
- There is a lot of overlap between the names of Toddler Module codes and other ADOS-2 modules, BUT the code descriptions on the Toddler Module are often different. **READ THE CODES CAREFULLY!**

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Coding Conventions

- 0 = Abnormality of type specified is NOT present (not necessarily "normal")
- 1 = Abnormality of type specified is present, but not sufficiently clear, severe, frequent, or marked to receive a code of 2
- 2 = Abnormality of type specified is definitely present and meets specific mandatory criteria
- 3 = Abnormality of type specified is present to a degree that may interfere with functioning
- 7 = Definite abnormality in general area of coding, but not of type specified
- 8 = Not applicable
- 9 = Unknown/missing

For reliability, the goal is 80% exact item agreement on protocol and algorithm, NOT including 2 vs. 3 as disagreements.

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Toddler Module Coding

- Coding is based on the behavioral samples obtained during the Toddler Module activities
- Code **ONLY** what you see *during* the administration
 - Behaviors observed before or after the Toddler Module can be incorporated into overall clinical impressions.
- **Code the behavior, interpret it later!**
 - Be careful about coding by diagnostic impression.
 - Code what you see.

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Toddler Module Coding (cont.)

Important Considerations

- Majority of examples vs. the best example
 - Read the specific instructions for each code!
- High-incidence vs. low-incidence behaviors
 - Expectations for frequency differ depending on the particular behavior (e.g., eye contact vs. giving, showing vs. repetitive behaviors).
- “Double” coding
 - It is not permissible to code the same aspect of the same behavior twice. You can code different aspects of the same behavior in different places (e.g., lining objects up, then peering at them).

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Calculating Reliability

- Goal: at least 80% interrater agreement or agreement with consensus on overall protocol and algorithm items
- Protocol Reliability =
$$\frac{\text{Number agree}}{\text{Total Number of Items}^*}$$
- Algorithm Reliability =
$$\frac{\text{Number agree on Algorithm}}{\text{Total Number of Algorithm Items}^*}$$

*Do not include items consensus-coded as 8 or 9 in numerator or denominator, unless you coded differently.

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Interpretation and Appropriate Use of ADOS-2 Toddler Module

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Algorithms

ALL children receive Toddler Module activities and codes

- All children between **12 and 20 months** of age
- Nonverbal children between **21 and 30 months** of age
- Verbal children between **21 and 30 months** of age



**All Younger/Older With
Few to No Words**



Older With Some Words

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Algorithms (cont.)

- Transfer selected scores from protocol to algorithm
- Convert scores (3 to 2; 7, 8, and 9 to 0)
- Combine algorithm domain total scores for Social Affect (SA) and Restricted and Repetitive Behavior (RRB)
- Combine SA + RRB totals = Overall Total
- Convert Overall Total to the ADOS-2 Range of Concern

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Ranges of Concern

- Used to inform overall clinical impression
 - **Little-or-no**
 - Few behaviors indicative of an ASD
 - **Mild-to-moderate**
 - Behaviors which may be consistent with an ASD
 - **Moderate-to-severe**
 - More strongly consistent with an ASD

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Algorithm Cutoffs and Ranges

All Younger/Older With Few to No Words	Older With Some Words
Cutoff = 12	Cutoff = 10
Ranges of concern: • Little-or-no: 0–9 • Mild-to-moderate: 10–13 • Moderate-to-severe: 14+	Ranges of concern: • Little-or-no: 0–7 • Mild-to-moderate: 8–11 • Moderate-to-severe: 12+

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Measuring Accuracy of Classifications: Sensitivity and Specificity

- **SENSITIVITY** = How many children with XX condition are accurately classified as having XX on the diagnostic measure
 - Sensitivity of ADOS-2: How many children **with a clinical diagnosis of ASD** score in either the ASD or the autism range
- **SPECIFICITY** = How many children who do NOT have XX condition are accurately classified as NOT having XX on the diagnostic measure
 - Specificity of ADOS-2: How many children **without a clinical diagnosis of ASD** score in the non-spectrum range

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Sensitivity and Specificity: Toddler Module*

Algorithm	ASD vs. NS	ASD vs. NS/TD
All Younger/Older With Few to No Words	(n = 35 vs. n = 34)	(n = 35 vs. n = 101)
Sensitivity	91%	91%
Specificity	91%	94%
Older With Some Words	(n = 24 vs. n = 11)	(n = 24 vs. n = 47)
Sensitivity	88%	88%
Specificity	91%	94%

*Algorithms validated for children with nonverbal mental age of at least 12 months. Use with discretion with children with nonverbal mental age < 12 months.

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RESEARCH ARTICLE

Replication study of ADOS-2 Toddler Module cut-off scores for autism spectrum disorder classification

Ji Su Hong^{1,2} | Vini Singh¹ | Luke Kalb^{1,3} | Ahlam Ashkar¹ | Rebecca Landa^{1,2}

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Participants

- Participants included 412 toddlers ages 12 to 30 months
 - 305 with ASD, 107 with non-ASD developmental or mental disorders
- Received an ASD diagnostic evaluation at an ASD specialty clinic
- **Sex:** 77.7% male
- **Race:** 49.5% White, 21.8% Black, 10.7% Asian
- **IQ:** Based on data from Mullen Scales of Early Learning (MSEL), 76% had Early Learning Composite (ELC; overall developmental skills) at the very low level

Source: Hong, J. S., Singh, V., Kalb, L., Ashkar, A., & Landa, R., 2021.

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Results

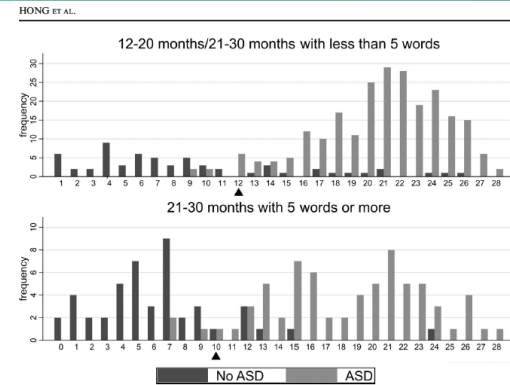
- Using recommended **cutoff score of 12** for toddlers ages 12–20 months and those ages 21–30 months with <5 words
 - Sensitivity of **98.31%** and a specificity of **75.41%**
- Using recommended **cutoff score of 10** for toddlers ages 21–30 months with 5 words or more
 - Sensitivity of **95.65%** and a specificity of **84.78%**
- Supports validity of the recommended cutoff scores in a much larger sample

Source: Hong, J. S., Singh, V., Kalb, L., Ashkar, A., & Landa, R., 2021.

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Frequency of Algorithm Score Between ASD vs. No-ASD Groups



Source: Hong, J. S., Singh, V., Kalb, L., Ashkar, A., & Landa, R., 2021.

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Methods

- Examined stability of the ADOS-2 Toddler Module scores and classifications across two time points
- Participants included 82 children identified through the First Words Project, a longitudinal study of a general population sample
 - Gender:** 78% male
 - Race:** 73.2% White, 14.6% Black, 2.4% Asian, 9.8% biracial
- Children who screened positive for ASD received comprehensive diagnostic evaluations at time 1 (between 15 and 24 months of age) and time 2 (at least 12 months later)

Source: Guthrie, W., Swineford, L. B., Nottke, C., & Wetherby, A. M., 2013.

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The Journal of Child Psychology and Psychiatry



Early diagnosis of autism spectrum disorder: stability and change in clinical diagnosis and symptom presentation

Whitney Guthrie, Lauren B. Swineford, Charly Nottke, Amy M. Wetherby

First published: 19 October 2012 | <https://doi.org/10.1111/jcpp.12008> | Citations: 214

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Results

- Clinical diagnosis confirming or ruling out ASD had a stability rate of 100%
- For individuals who received a deferred diagnosis (17%), stability was poor
- The Toddler Module research classification cutoffs showed excellent discrimination
- Time-1 Toddler Module scores significantly predicted time-2 clinical diagnosis

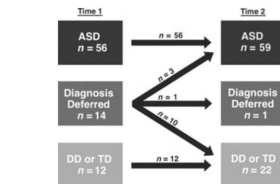


Figure 1 Stability of clinical diagnosis

Source: Guthrie, W., Swineford, L. B., Nottke, C., & Wetherby, A. M., 2013.

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Results (cont.)

Table 3 Agreement between Time 1 classification and Time 2 diagnosis/classification

		T2 Clinical diagnosis		T2 ADOS Diagnostic classification		
		ASD n (% within classification)	Non-ASD	Autism n (% within classification)	ASD	Nonspectrum
T1 ADOS-T ranges of concern	Little to no (n = 12)	2 (17%)	10 (83%)	1 (8%)	1 (8%)	10 (83%)
	Mild to moderate (n = 26)	15 (58%)	11 (42%)	13 (50%)	4 (15%)	9 (35%)
	Moderate to severe (n = 44)	42 (95%)	2 (5%)	37 (84%)	5 (11%)	2 (5%)
		$\chi^2(2) = 32.84, p < .001$		$\chi^2(4) = 32.84, p < .001$		
T1 ADOS-T diagnostic classification	Nonspectrum (n = 25)	7 (28%)	18 (72%)	7 (28%)	2 (8%)	16 (64%)
	ASD (n = 57)	52 (91%)	5 (9%)	44 (77%)	8 (14%)	5 (9%)
		$\chi^2(1) = 34.43, p < .001$		$\chi^2(2) = 27.98, p < .001$		

ADOS-T, Autism Diagnostic Observation Schedule – Toddler Module; ASD, autism spectrum disorder.

Source: Guthrie, W., Swineford, L. B., Nottke, C., & Wetherby, A. M., 2013.

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Important Caveats About the Sensitivities and Specificities of the ADOS-2

- The ADOS-2 Toddler Module sensitivity and specificity values are based on data from examiners who are research reliable.
- The sensitivity and specificity of the ADOS-2 Toddler Module is likely to be lower when used by examiners who are not research reliable.
- All examiners (research reliable or not) must take care to interpret the ADOS-2 Toddler Module results in the context of all other information.
- Examiners should also consider differences between their clinical population and the validation samples in terms of medical complexity, cultural diversity, behavioral issues, etc.

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ADOS-2 Comparison Score (i.e., Calibrated Severity Score) for Toddler Module

Purpose

- To indicate level of autism-related symptomatology, as observed during the ADOS-2, as compared with children diagnosed with ASD who are of the same chronological age and language level.
- To help interpret an individual's changes over time (i.e., across age and modules).
- Calibrated Severity Score (CSS) available for the total algorithm scores as well as SA and RRBs domains.
- CSS conversion table available in Esler et al., 2015.

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CSS Conversion Table: Toddler Module

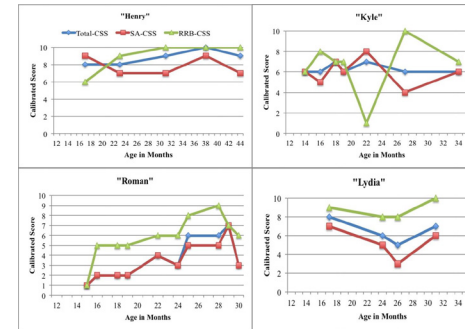
Toddler module concern range	CSS	Raw totals					
		Overall total		SA domain		RRB domain	
		12-20/NV	SW 21-30	12-20/NV	SW 21-30	12-20/NV	SW 21-30
Little-to-no	1	0-2	0-3	0-2	0-1	0	0
	2	3-5	4-6	3-4	2-3	—	—
	3	6-9	7	5-6	4-5	—	—
Mild-to-moderate	4	10-11	8-9	7-9	6-8	—	—
	5	12-13	10-11	10	9-10	1-2	1
Moderate-to-severe	6	14-16	12	11-12	11	3	2
	7	17-18	13-15	13-14	12-13	4	3
	8	19-21	16-17	15-16	14-15	5	4
	9	22-23	18-20	17-18	16-18	6	5
	10	24-28	21-28	19-20	19-22	7-8	6

Source: Esler, A. N., et al., 2015.

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Using the CSS

Tracking Individual Toddler Trajectories With the CSS

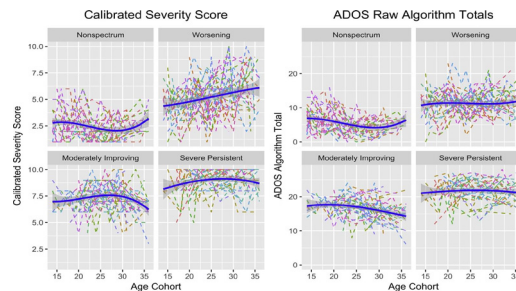


Source: Esler, A. N., et al., 2015.

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Using the CSS (cont.)

CSS are less affected by developmental levels, allowing the comparison of scores across time for children who move from one module to another.

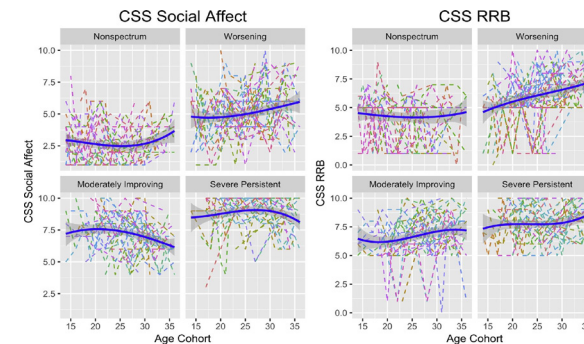


Source: Kim et al., 2018.

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Using the CSS (cont.)

Social affect and RRBs may show varying trajectories.



Source: Kim et al., 2018.

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What the Comparison Score Is NOT

- It does **NOT** represent the child's ASD symptoms or overall functioning relative to a typically developing reference group.
 - It is in reference to children with ASD of the same age and language level.
- It is **NOT** based on a population or epidemiological sample.
 - It is derived from a large clinical/research sample.
- It does **NOT** measure daily functioning or adaptive behavior.
 - It measures ASD-specific symptoms in relation to children of the same age and language level in the context of the ADOS-2.

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Incorporating ADOS-2 Toddler Module Results Into Evaluation Reports

- Provide description of ADOS-2 Toddler Module (why administered, types of activities)
- Report diagnostically relevant behaviors observed during the ADOS-2 Toddler Module (social–communication and repetitive behaviors)
- Do not report raw scores; use Ranges of Concern
- If validity criteria are met and it is otherwise appropriate to do so, may report classification and/or comparison scores

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ADOS-2 Summary of Findings Example

- The ADOS-2 Toddler Module results in three ranges of concern (little-or-no, mild-to-moderate, moderate-to-severe) for Autism Spectrum Disorder. On this administration of the ADOS-2 Toddler Module, [name's] scores fell into the moderate-to-severe range of concern.
- The ADOS-2 comparison score measures an individual's level of ASD-related behaviors compared to other individuals with ASD of the same age and expressive language level. On this administration of the ADOS-2 Toddler Module, [name] received a Comparison Score of 8, indicating a high level of ASD-related behaviors exhibited during the ADOS-2.

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ADOS-2 Description Toddler Module

- As part of this evaluation, [child's name] was given the Autism Diagnostic Observation Schedule, Second Edition (ADOS-2) Toddler Module. The ADOS-2 Toddler Module is a structured observation designed to elicit social and communication behaviors in toddlers with little to no speech. The Toddler Module involves structured and unstructured tasks, during which the examiner engages in a variety of interactions with the child. The Toddler Module includes opportunities for adult-led interactions, such as having a pretend bath time for a doll, playing with bubbles and balloons, and imitating actions with objects, as well as opportunities to observe the child in spontaneous play during free play and snack activities.

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ADOS-2 Social Affect Summary Young Child

- In the area of social communication, Eve did not consistently use eye contact. She used word approximations, but she did not consistently direct these vocalizations to others. More frequently, she took the examiner's hand and placed it on objects of interest. In one instance, Eve used an approximation of a point to request a snack. When the examiner pointed to a toy bunny, Eve responded by turning her head and saying, "bu." She also responded by making eye contact and smiling when her mother called her name. Eve seemed to enjoy many of the ADOS-2 activities. She did not generally share her enjoyment with others, though she did laugh briefly when the examiner tickled her.

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ADOS-2 RRB Summary Young Child

- John enjoyed several of the ADOS-2 activities, including make-believe play and the pretend bath time activity. During make-believe play with a family of dolls, John placed the dolls on the toy sofas and had a doll carry miniature food. While having a pretend bath time for a doll, John participated by singing along. He particularly liked it when the examiner pretended that the soap went in the baby's eyes, so he imitated this action several times. John also tended to play with toys or objects in a repetitive way and to focus on the sensory aspects of different objects. For example, when using a bubble blower, John visually inspected the spinning fan and peered at it out of the corner of his eye.

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Was my ADOS valid?

- Things you can more or less control:
 - Standardized administration and materials
 - Caregiver present for T/1/2
 - Adequate time (at least 30 minutes but no more than 60 minutes)
- Things you cannot necessarily control:
 - Child behavior, emotional state during ADOS
 - Child developmental level
 - Child motor skills, vision, hearing
 - COVID-19 protocols for use of PPE, etc.

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What if you feel it is not valid?

- An attempt was made to administer the Autism Diagnostic Observation Schedule, Second Edition (ADOS-2) Module T to [name]; however, all criteria for ADOS-2 validity were not met, specifically in the area of [insert validity concern]. Instead, qualitative observations can be made from several play-based tasks that were administered.
- (Proceed describing what behaviors were observed related to language, communication, reciprocal social interaction, stereotyped behaviors, play skills, and E codes.)

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Toddler Module Summary

- The ADOS-2 Toddler Module is a semi-structured standardized assessment that creates contexts in which to observe ASD-related symptoms and behaviors in young children with minimal expressive language abilities who are 12–30 months of age.
- It helps to clarify whether a child is demonstrating behaviors associated with an eventual diagnosis of ASD.
- It provides a useful way of quantifying risk for ASD and identifying need for continued monitoring and follow up.

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Toddler Module Summary (cont.)

- The Toddler Module is one component of a comprehensive evaluation. Evaluation should also include:
 - Interview with parent/caregiver to obtain full developmental history
 - Cognitive and language testing
 - Medical assessment and hearing screening
 - Observation in and/or reports about behavior in relevant environments (e.g., home, daycare)
- Clinician may be asked to consider ASD versus global developmental delay and/or language delay.
- Even with such a comprehensive evaluation, a formal diagnosis may not yet be appropriate for a very young child.
 - Monitoring with repeat assessment is necessary for some toddlers.

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Where to go from here?

- Required preparation for *clinical* and *research* use:
 - Read the Manual and protocol carefully.
 - Practice *extensively* with a variety of individuals, including typically developing children and those with ASD and non-ASD diagnoses.
 - “Shadow” a seasoned ADOS-2 examiner.
 - Participate in refresher trainings and other development opportunities, such as Global Autism Interactive Network (GAIN).
- Required preparation for *research* use (also ideal for clinical use):
 - Attend ADOS-2 Advanced/Research Training.
 - Establish research reliability.

ADOS-2 Toddler Module should only be used by people who frequently assess toddlers with ASD.

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Further Practice: GAIN! Global ADOS-2 Intervention Network

- GAIN is a professional development program designed to improve proficiency on the ADOS-2. GAIN is a virtual, live-training webinar hosted by ADOS-2 trainers.
- GAIN meets the fourth Friday of each month; it is divided into quarters, each focused on a different module of the ADOS-2.
- Additional sessions have been added to accommodate different time zones.
- Join GAIN here: <https://psychiatry.weill.cornell.edu/education-training/autism/gain>
- Email the GAIN training coordinator at GAIN@med.cornell.edu with any questions.

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Contact Information

- WPS
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- Designated training sites for the ADOS-2 and ADI-R:
 - UCSF Center for ASD and NDDs
<https://autism.ucsf.edu/training>
 - Center for Autism and the Developing Brain (CADB)
<https://psychiatry.weill.cornell.edu/education-training/autism-training>
 - University of Minnesota (UMN)
<https://med.umn.edu/pediatrics/divisions/clinical-behavioral-neuroscience/education/autism-training>
 - Rutgers University
<https://rcaas.rutgers.edu/about/professional-training>
 - SpectrumKU (Korea University)
<https://spectrumku.com/main/main.html>

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ADOS-2 Additional Resources and Explainers

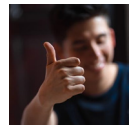
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Different Types of Gestures (as Defined by the ADOS-2)

REMEMBER: To be considered a gesture, it must be communicative; in general, it is not considered a gesture if the participant is holding something in their hand.

- **Descriptive:** pantomime actions used when trying to describe or represent an action, object, or event for communicative purposes (e.g., size, fishing cast, flying, blow to request blowing a balloon, brushing teeth)
- **Conventional:** have a standard social meaning within a particular culture; learned or modeled (e.g., thumbs up, wave, high five, clap, nodding/shaking head)
- **Instrumental:** intended to achieve a goal (e.g., beckoning someone to come here, communicative reach)

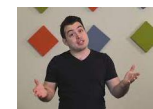


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Different Types of Gestures (as Defined by the ADOS-2) (cont.)

- **Informational:** to convey specific information (e.g., holding up two fingers to mean "two")
- **Emotional:** express an emotion (e.g., shaking fists to mean angry, putting one's hand to one's mouth and gasping to indicate shock)
- **Emphatic:** hand movements (beats) that are integrated into speech to provide emphasis as one is talking (Module 4 only)



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Facial Expressions

- Look for:
 - Range
 - Directedness
- Can be subtle, especially with older participants
- Need to be directed toward people
 - When eye contact is already low, look for other signs of directedness

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Social Overtures vs. Social Response

Social Overtures	Social Responses
- Behavior <u>initiated</u> by the examinee - Any attempt to get or maintain the examiner's attention for any purpose	- Behavior <u>in response</u> to another person OR to the situation - Absence of an expected behavior (such as ignoring examiner) - Consider: - Response to presses - Response to social context - Absence of appropriate responses

Remember: Consider both quality and frequency

(see ADOS-2 Manual, p. 19)

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Behaviors to Consider in Social Overtures and Responses

	Social Overtures	Social Responses	Social Overture or Social Response
Verbal	- Offering information during a conversation after a pause - Asking questions - Greeting - Requesting - Commenting - Asking for clarification	- Answering questions - Refusing/protesting - Making non-lexical conversational sounds - Making undirected vocalizations	- Intonation - Stereotyped speech - Bringing up preoccupation - Seeking/offering comfort - Sharing enjoyment (verbal) - Screaming/yelling
Nonverbal	- Initiating by showing a toy or object - Giving a toy or object without prompt - Using spontaneous joint attention to direct attention - Moving closer to indicate enjoyment/continuation - Requesting (nonverbal) - Sharing without prompt - Initiating a game	- Nodding as someone talks - Modulating eye contact - Changing facial expressions - Moving away/withdrawing - Turning to name - Responsive social smile - Reciprocating in a game	- Eye contact - Being overly aggressive or impulsive - Changing facial expressions - Joining an activity (physically) - Pointing/gesturing - Moving another's hand or body - Odd affect/laughing

Could be either (or neither): Context determines whether the behavior is an overture or response

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Amount of Reciprocal Social Communication

Social Overtures	Social Responses	Amount of Reciprocal Social Communication
- Behavior <u>initiated</u> by the examinee - Has communicative intent - Attempt to get or maintain person's attention	- Behavior <u>in response</u> to another person or the situation - Absence of an expected behavior (such as ignoring examiner) - Consider: - Response to presses - Response to social context - Absence of appropriate responses	- Behaviors that are in the service of the social interaction - Can be verbal or nonverbal - Can be very brief - Can be sophisticated or "low level" - E.g., saying "thank you" or "bless you," "mm hmm" and/or nodding along with what you are saying, checking in with eye contact, trying to help when you look like you need it - In Module 2, can be directed to examiner or parent

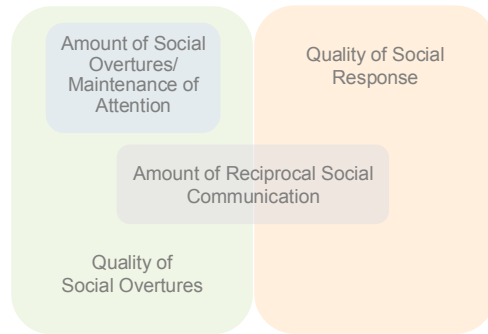
If any of these lead to, or are for the benefit of

The social interaction

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How They Fit Together



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Module 1: Code Explainer Integration of Gaze and Other Behaviors During Social Overtures

3 Different Components

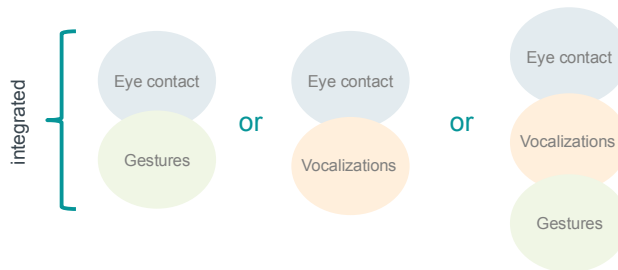


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Integration of Gaze and Other Behaviors During Social Overtures

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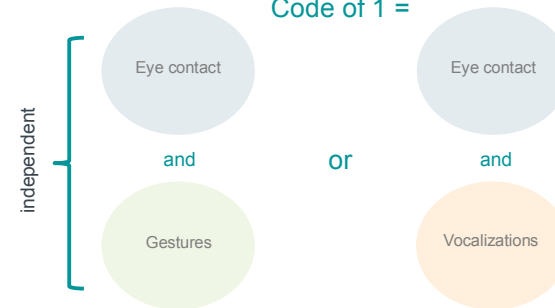
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Integration of Gaze and Other Behaviors During Social Overtures (cont.)



Code of 1 =



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Integration of Gaze and Other Behaviors During Social Overtures (cont.)



Code of 2 =



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Integration of Gaze and Other Behaviors During Social Overtures (cont.)



Code of 3 =

Nothing

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